Classwork 01// Matrix Calculus: Jacobian

Student Name: ______________________________ **Points:** 30

A Number: ____

1. Derive the Jacobian of identify function $\mathbf{f}(\mathbf{x}) = \mathbf{x}$, i.e. $f_i(\mathbf{x}) = x_i$ that has n functions and each function has n parameters held in a single vector $\mathbf{x}$. **(10 Points)**
2. Under what conditions, are the off-diagonal elements of a Jacobian zero? **(10 Points)**
3. Compute gradient of $y = sum(\mathbf{x})$, i.e. ∇y, considering the function inside the summation as $f_i(\mathbf{x}) = x_i$. Is it a vertical vector or a horizontal vector as per convention used in the Lecture notes? **(10 Points)**

Solution

1. Consider $\mathbf{y} = \mathbf{f}(\mathbf{x})$ for shorthand notation. As the function given in the problem is an identity function, so

$$\begin{aligned} y_1 &= f_1(\mathbf{x}) = x_1 \\ y_2 &= f_2(\mathbf{x}) = x_2 \\ &\vdots \end{aligned}$$

So we can write Jacobian as

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial \mathbf{x}} \\ \frac{\partial f_2(\mathbf{x})}{\partial \mathbf{x}} \\ \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial \mathbf{x}} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \frac{\partial f_1(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f_1(\mathbf{x})}{\partial x_n} \\ \frac{\partial f_2(\mathbf{x})}{\partial x_1} & \frac{\partial f_2(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f_2(\mathbf{x})}{\partial x_n} \\ & & \dots & \\ \frac{\partial f_m(\mathbf{x})}{\partial x_1} & \frac{\partial f_m(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_n} \end{bmatrix}$$

since $\frac{\partial x_i}{\partial x_j} = 0$ for $j \neq i$ because f_1 only has x_1 term, f_2 only has x_2 term and so on. Hence,

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial x_1}{\partial x_1} & 0 & \cdots & 0 \\ 0 & \frac{\partial x_2}{\partial x_2} & \cdots & 0 \\ & & \cdots & \\ 0 & 0 & \cdots & \frac{\partial x_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ & & \cdots & \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I$$

I is the identity matrix.

2. When scalar functions are depenedent on one variable (feature) only such that f_1 only has x_1 term, f_2 only has x_2 term and so on, then the off diagonal elements of a Jacobian will be zero.

3. $y = sum(\mathbf{x})$, and the function inside the summation is just $f_i(\mathbf{x}) = x_i$ and then the gradient is

$$\begin{aligned} \nabla y &= \begin{bmatrix} \sum_i \frac{\partial f_i(\mathbf{x})}{\partial x_1} & \sum_i \frac{\partial f_i(\mathbf{x})}{\partial x_2} & \cdots & \sum_i \frac{\partial f_i(\mathbf{x})}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \sum_i \frac{\partial x_i}{\partial x_1} & \sum_i \frac{\partial x_i}{\partial x_2} & \cdots & \sum_i \frac{\partial x_i}{\partial x_n} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial x_1}{\partial x_1} & \frac{\partial x_2}{\partial x_2} & \cdots & \frac{\partial x_n}{\partial x_n} \end{bmatrix} = [1 \quad 1 \quad \cdots \quad 1]] = \vec{1}^\top \end{aligned}$$

which is a horizontal vector.